Swiss TPH

Quarto @ Swiss TPH

Usage of Swiss TPH Quarto Templates

Zhihan Zhu (Research Informatics)

Agenda

What is Quarto?
Swiss TPH Quarto Templates
Use Case – Application on PARVAL Project

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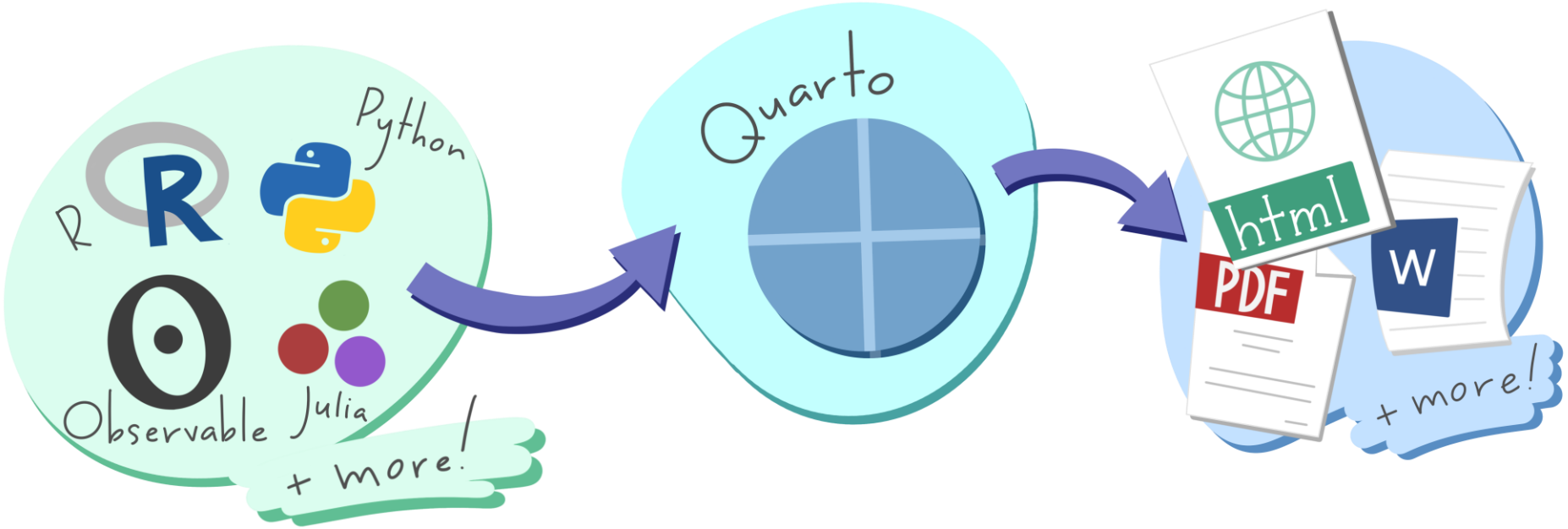
What is Quarto?

What is Quarto?



Artwork by Allison Horst.

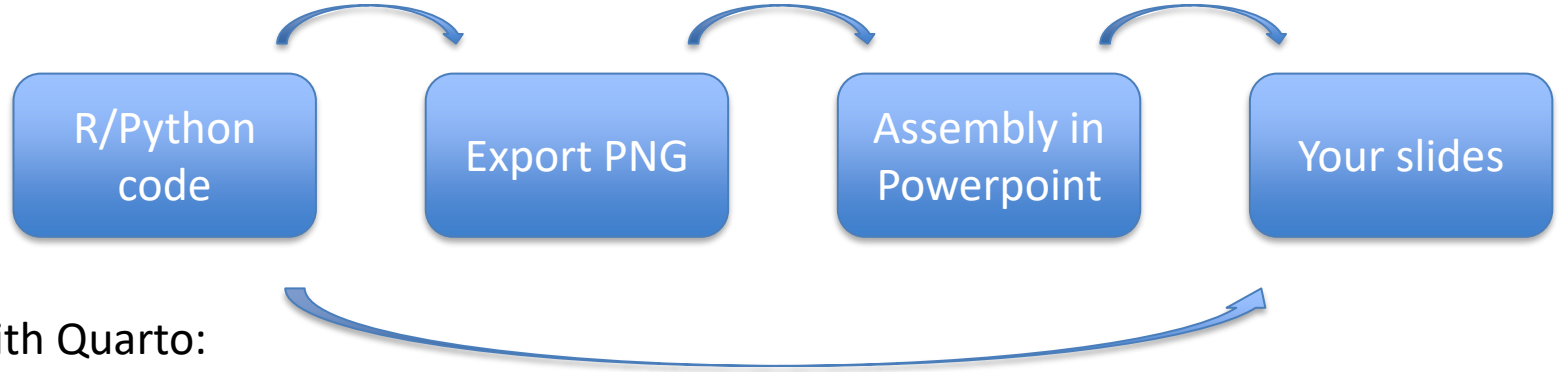
Why Quarto?

Multi-language	Quarto is not tied to R like R Markdown does. It supports multiple languages including **R, Python, Julia, and Observable**.
High compatibility with existing work	You can render most existing **R Markdown** documents and **Jupyter notebooks** with Quarto, keeping the switching cost low.
Long-term investment in reproducible research	Quarto is positioned as a major long-term platform investment for reproducible scientific publishing, while staying compatible with existing formats and user workflows.

Application Scenario

Usual preparation:

Complexity when changes occur
Prone to making mistakes



With Quarto:

Reproducible anytime
Convenient and fast

Swiss TPH

Swiss TPH Quarto Templates

Why use Swiss TPH Quarto templates?

- 🎨 Maintain Swiss TPH corporate identity: Fonts, colors, layouts, and logos are predefined and consistent across outputs.
- ⏱️ Save time: No need to manually format documents or presentations—focus on your content instead.
- 🔁 Ensure reproducibility: The same source file can be re-rendered with updated data or text at any time.

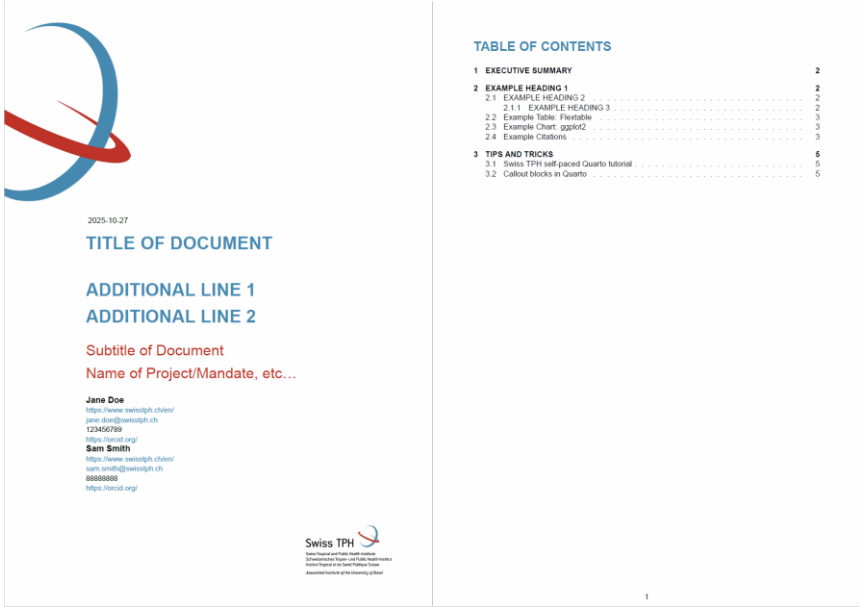
What are available?

For now, Swiss TPH Quarto templates are available in these formats:

- PDF report (Latex and Typst)
- PDF letterhead (Latex)
- Word
- Powerpoint
- Revealjs

Contributors: Hélène Langet, Zhihan Zhu

Swiss TPH Quarto template gallery



2025-10-27

TITLE OF DOCUMENT

ADDITIONAL LINE 1
ADDITIONAL LINE 2

Subtitle of Document
Name of Project/Mandate, etc…

Jane Doe
https://www.swisstph.ch/en/
jane.doe@swisstph.ch
123456789
https://orcid.org/
Sam Smith
https://www.swisstph.ch/en/
sam.smith@swisstph.ch
88888888
https://orcid.org/

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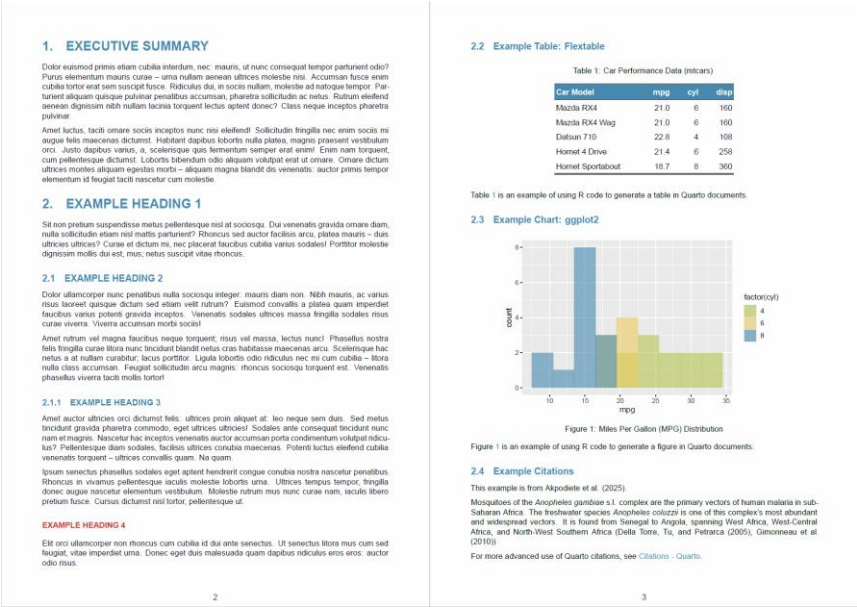
PDF(Latex) cover page and table of content

Swiss TPH Quarto template gallery



1. EXECUTIVE SUMMARY

Dolor euismod primis etiam cubilia interdum, nec: mauris, ut nunc consequat tempor parturient odio? Purus elementum mauris curae – urna nullam aenean ultrices molestie nisi. Accumsan fusce enim cubilia tortor erat sem suscipit fusce. Ridiculus dui, in sociis nullam, molestie ad natoque tempor. Parturient aliquam quisque pulvinar penatibus accumsan, pharetra sollicitudin ac netus. Rutrum eleifend aenean dignissim nibh nullam lacinia torquent lectus aptent donec? Class neque inceptos pharetra pulvinar.

Amet luctus, taciti ornare socios inceptos nunc nisi eleifend! Sollicitudin fringilla nec enim socios mi augue felis maecenas dictumst. Habitant dapibus lobortis nulla platea, magnis praesent vestibulum orci. Justo dapibus varius, a, scelerisque quis fermentum semper erat enim! Enim nam torquent, cum pellentesque dictumst. Lobortis bibendum odio aliquam volutpat erat ut ornare. Ornare dictum ultrices montes aliquam egestas morbi – aliquam magna blandit dis venenatis: auctor primis tempor elementum id feugiat taciti nascetur cum molestie.

2. EXAMPLE HEADING 1

Sit non pretium suspendisse metus pellentesque nisl at sociosqu. Dui venenatis gravida ornare diam, nulla sollicitudin etiam nisl mattis parturient? Rhoncus sed auctor facilisis arcu, platea mauris – duis ultricies ultrices? Curae et dictum mi, nec placerat faucibus cubilia varius sodales! Porttitor molestie dignissim mollis dui est, mus; netus suscipit vitae rhoncus.

2.1 EXAMPLE HEADING 2

Dolor ullamcorper nunc penatibus nulla sociosqu integer: mauris diam non. Nibh mauris, ac varius risus laoreet quisque dictum sed etiam velit rutrum? Euismod convallis a platea quam imperdiet faucibus varius potenti gravida inceptos. Venenatis sodales ultrices massa fringilla sodales risus curae viverra. Viverra accumsan morbi socis!

Amet rutrum vel magna faucibus neque torquent; risus vel massa, lectus nunc! Phasellus nostra felis fringilla curae litora nunc tincidunt blandit netus cras habitasse maecenas arcu. Scelerisque hac netus a at nullam curabitur; lacus porttitor. Ligula lobortis odio ridiculus nec mi cum cubilia – litora nulla class accumsan. Feugiat sollicitudin arcu magnis: rhoncus sociosqu torquent est. Venenatis phasellus viverra taciti mollis tortor!

2.1.1 EXAMPLE HEADING 3

Amet auctor ultricies orci dictumst felis: ultrices proin aliquet at: leo neque sem duis. Sed metus tincidunt gravida pharetra commodo, eget ultrices ultricies! Sodales ante consequat tincidunt nunc nam et magnis. Nascetur hac inceptos venenatis auctor accumsan porta condimentum volutpat ridiculus? Pellentesque diam sodales, facilisis ultrices conubia maecenas. Potenti luctus eleifend cubilia venenatis torquent – ultrices convallis quam. Na quam.

Ipsum senectus phasellus sodales eget aptent hendrerit congue conubia nostra nascetur penatibus. Rhoncus in vivamus pellentesque iaculis molestie lobortis urna. Ultrices tempus tempor, fringilla donec augue nascetur elementum vestibulum. Molestie rutrum mus nunc curae nam, iaculis libero pretium fusce. Cursus dictumst nisl tortor, pellentesque ut.

EXAMPLE HEADING 4

Elit orci ullamcorper non rhoncus cum cubilia id dui ante senectus. Ut senectus litora mus cum sed feugiat, vitae imperdiet urna. Donec eget duis malesuada quam dapibus ridiculus eros eros: auctor odio risus.

2

2.2 Example Table: Flextable

Table 1: Car Performance Data (mtcars)

Car Model	mpg	cyl	disp
Mazda RX4	21.0	6	160
Mazda RX4 Wag	21.0	6	160
Datsun 710	22.8	4	108
Hornet 4 Drive	21.4	6	258
Hornet Sportabout	18.7	8	360

Table 1 is an example of using R code to generate a table in Quarto documents.

2.3 Example Chart: ggplot2

Figure 1: Miles Per Gallon (MPG) Distribution

Figure 1 is an example of using R code to generate a figure in Quarto documents.

2.4 Example Citations

This example is from Akpodiete et al. (2025).

Mosquitoes of the *Anopheles gambiae* s.l. complex are the primary vectors of human malaria in sub-Saharan Africa. The freshwater species *Anopheles coluzzii* is one of this complex's most abundant and widespread vectors. It is found from Senegal to Angola, spanning West Africa, West-Central Africa, and North-West Southern Africa (Della Torre, Tu, and Petrarca (2005), Gimonneau et al. (2010))

For more advanced use of Quarto citations, see Citations - Quarto.

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PDF(Latex) headings, text, tables and plots

Swiss TPH Quarto template gallery

3.2 Callout blocks in Quarto

You can create colourful callout blocks in Quarto very easily.

> **Note**
>
> Note that there are five types of callouts, including: `note`, `warning`, `important`, `tip`, and `caution`.

See Callout Blocks for more details.

> **Swiss TPH Callout Box**
>
> Or you can create more flexible callout blocks with LaTeX. Just like this one with Swiss TPH lime green.

References

Akpodiete, Nwamaka Oluchukwu, Bianca Carlos, Kamila Voges, Bruno Tinoco Nunes, Jayme Augustus Souza-Neto, Florian Noulin, Daniel Tonge, Wan Fatma Zuharah, and Frédéric Tripet. 2025. "Improvement of Water Quality for Mass Anopheline Rearing: Dynamics of Larval Tray Bacterial Communities Under Different Water Treatments Revealed by 16S Ribosomal RNA Amplicon Sequencing." *Journal of Applied Microbiology* 136 (5): lxaf110. https://doi.org/10.1093/jambio/lxaf110.

Della Torre, Alessandra, Zhijian Tu, and Vincenzo Petrarca. 2005. "On the Distribution and Genetic Differentiation of Anopheles Gambiae s.s. Molecular Forms." *Insect Biochemistry and Molecular Biology* 35 (7): 755–69. https://doi.org/10.1016/j.ibmb.2005.02.006.

Gimonneau, Geoffrey, Jérémy Bouyer, Serge Morand, Nora J. Besansky, Abdoulaye Diabate, and Frédéric Simard. 2010. "A Behavioral Mechanism Underlying Ecological Divergence in the Malaria Mosquito Anopheles Gambiae." *Behavioral Ecology* 21 (5): 1087–92. https://doi.org/10.1093/beheco/arq114.

PDF(Latex) callout boxes, references, and more…

Swiss TPH

Use case of Swiss TPH Quarto templates

Reports for PARVAL project

PARVAL is a study aims to assess pesticide exposure in school children in Valais, Switzerland.

A task at the end of the project is to deliver personalized letters for study participants.

Participants in the task of creating reports:

Zhihan Zhu (Research Informatics)

Lucienne Zinsstag (PARVAL Study Team)

Aline Berweger (PARVAL Study Team)

Hélène Langet (Research Informatics)

Samuel Fuhrimann (PARVAL Study Lead)

Screenshots for PARVAL project reports

Reports for PARVAL project

260 reports with one click…

- Swiss TPH logo.
- Letter head and signature.
- Links with customized color (Swiss TPH blue!)
- Personalized icons and highlight box.
- **Reproducible personalized tables and plots for different individuals.**
- Received good feedback.

Screenshots for PARVAL project reports

Meet the team

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Resources

- Swiss TPH Quarto Training
- Swiss TPH Toolbox, including Quarto templates
- Quarto - Get Started